FULL NAME: ______________________________ **BLOCK:** ____________

Logarithms + DOK-3 Spine: q15 (exp↔log inverse via graph)

Lesson 6-3 · Quick · Teacher Edition

OBJECTIVES

Math Objective	Evaluate logarithms using the definition $\log_b x = y \Leftrightarrow b^y = x$, distinguish common (log) vs natural (ln) logs, and use the graph of $y = 3^x$ to estimate logarithmic values through the exp↔log inverse relationship.
Language Objective	Explain the inverse relationship between exponential and logarithmic equations using "undo" language.
Essential Understanding	A logarithm answers the question: what exponent? – logs undo exponentials. If I know how to graph $y = 3^x$, I can use that same picture to estimate $\log_3 50$ because logarithms and exponentials are inverses.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency evaluating logarithms; use the exp↔log inverse relationship graphically (Topic 6 Form B prep: q15 covers Q7, Q10, Q11; q17 covers Q13).
Essential Question	If I know how to graph $y = 3^x$, how can I use that same picture to estimate $\log_3 50$ — and why does this work?
Materials	Student packet with printed $y = 3^x$ graph ($x \in [-1, 4.5]$, $y \in [0, 85]$, gridlines). Teacher copy adds dashed $y = 50$ line and read-off annotation. [GRAPH PLACEHOLDER — needs_cleanup=YES, see Visuals Checklist].

[PIN] PRE-FLIGHT — SINGLE-PERIOD COMPRESSED LESSON + DOK-3 SPINE q15

This is a single-period quick treatment of Lesson 6-3. ONE example modeled in Launch (Ex 3: Evaluate Logarithms). DOK-3 spine q15 drives Explore.
q15: Use the graph of $y = 3^x$ to estimate $\log_3 50$. Students find $y = 50$ on the graph and read off the x-value. Since $3^3 = 27$ and $3^4 = 81$, answer is between 3 and 4, closer to 3.5. Formally $\log_3 50 \approx 3.56$.
Pair-share: why does this work? (Because $\log_3 50 = x$ means $3^x = 50$ — logs and exp are inverses.)
VISUAL REQUIRED: Student packet needs printed $y = 3^x$ graph ($x \in [-1, 4.5]$, $y \in [0, 85]$, gridlines).
Teacher copy: same graph + dashed $y = 50$ line + read-off annotation. See Visuals Checklist — needs_cleanup=YES.
PT#58 dropped as spine — not Form-B-aligned. q15 rehearses 3 of 8 Form B items through exp↔log inverse relationship. The continuous-growth modeling of PT#58 will live in a future lesson if the class picks up 6-6+.
Pace: Do Now (7) → Launch (8) → Explore q15 (15 min) + q52 (4) + q53 (3) + q54 (3) → Share (5) → Exit (5). ≈ 48 min. Buffer in q53+q54.
45-min Wed F variant: cut q53 + q54 from Explore. NEVER cut q15.

Tag: bridge to logarithms launch

Do Now **DOK:** 1–2 **Minutes:** 7

Teacher does

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student on q14 error analysis.
- Pull sheets at 7 min.

Students do

- Error analysis on $e^t = 98$ (q14).
- Reason about $\log_4 x < 0$ (q16).
- Sort vocab: common vs. natural log (q3).

Questions to ask

- What is the student's error in q14? What step undoes the exponential?
- For q16: what base-4 power gives a value less than 1?
- What's the base for log x vs. ln x?

Adult role

Solo teacher: circulate silently. No hints on q14 — let pairs argue.

Do Now #1 — Error Analysis (q14) (6-3-savvas-q14)

Error Analysis Describe and correct the error a student made in solving an exponential equation.

$$\begin{aligned} 16e^t &= 98 \\ e^t &= 6.125 \\ 6.125t &= \ln e \\ t &= \frac{\ln e}{6.125} \end{aligned}$$

[CHECK] ANSWER KEY / ANTICIPATED TALK

Answer key (from source): The student did not convert to logarithmic form correctly. The solution should be $t = \ln 6.125$.

Do Now #2 — Reasoning (q16) (6-3-savvas-q16)

Generalize For what values of x is the expression $\log_4 x < 0$ true?

[CHECK] ANSWER KEY / ANTICIPATED TALK

Answer key (from source): $\{x \mid 0 < x < 1\}$

Do Now #3 — Vocabulary (q3) (6-3-savvas-q3)

Vocabulary Explain the difference between the common logarithm and the natural logarithm.

[CHECK] ANSWER KEY / ANTICIPATED TALK

(verify before class)

Tag: teacher think-aloud Example 3 (Evaluate Logarithms)

Launch **DOK:** 1–2 **Minutes:** 8

Teacher does

- Project Example 3. Think aloud: $\log_b x = y \Leftrightarrow b^y = x$.
- Walk through $\log_5 125 = 3$ (ask: $5^? = 125$).
- Walk through $\log_{1/4} 16 = -2$ (negative exponent — flag this!).
- Walk through $\log_3 0$: Is there any power of 3 that gives 0? No – undefined.
- Walk through $\log_2 2^8 = 8$ using the identity $\log_b(b^k) = k$.
- Flag Rules card — students mark it on student packet.
- 30-sec pause: Turn to partner: rewrite $\log_4 64$ using the definition.
- Do NOT solve Practice items — release at minute 15.

Students do

- Watch Example 3 silently. Copy the log ↔ exp switch.
- Note: $\log_b 0$ is undefined (no real answer).
- Mark the Rules card on their student packet.
- During pause: rewrite $\log_4 64$ with partner.

Questions to ask

- $\log_5 125 = ?$ — what power does 5 need?
- $\log_{1/4} 16 = ?$ — why is the answer negative?
- Why is $\log_3 0$ undefined?
- Use the identity: $\log_2 2^8 = ?$

Adult role
Project Example 3 only. Release promptly at minute 15.

[MIC] TEACHER SCRIPT

1. 'A logarithm asks: what exponent does b need to produce x?'
2. 'log_5 125: what power of 5 is 125? $5^1 = 5$, $5^2 = 25$, $5^3 = 125$. Answer: 3.'
3. 'log_{1/4} 16: tricky — $(1/4)^? = 16$. $(1/4)^{-2} = 4^2 = 16$. Answer: -2.'
4. 'log_3 0: no power of 3 ever reaches 0 — UNDEFINED.'
5. 'log_2 2^8: identity $\log_b(b^k) = k$. Answer: 8. No work needed.'
6. 'For today's DOK-3 spine: on the graph of $y = 3^x$, we can read $\log_3 50$ directly — because exp and log are inverses, the x-value IS the log.'
7. Release to Explore.

Tag: TPS + DOK-3 driver

Explore **DOK:** 2–3 **Minutes:** 25

Teacher does

- 4 laps across 25 min.
- Lap 1 (10 min): [STAR] q15 DOK-3 spine. Press pairs: 'Where on the graph is $y = 50$? What x-value does that correspond to?' Do NOT give the answer.
- After 10 min: pair presentations (5 min). One pair explains: 'Why does reading the x-value on the exp graph give you the log?'
- Lap 2 (4 min): q52. Press: 'Which log rule applies here?'
- Lap 3 (3 min): q53. Press conversion between log and exp forms.
- Lap 4 (3 min): q54. Press asymptote and domain reasoning.
- Then q17: press 'Is $\log_3(1/27) = -3$ correct? Use the definition to check.'
- 45-min variant: skip q53 and q54. Budget extra time on q15.

Students do

- 5 items in order: q15 → q52 → q53 → q54 → q17.
- For q15: use the printed $y = 3^x$ graph. Find $y = 50$, read the x-value.
- Pair-share for q15: explain WHY reading the x-value gives $\log_3 50$.
- q17: verify or refute $\log_3(1/27) = -3$ using the definition.
- Justify each step using sentence frame.

Questions to ask

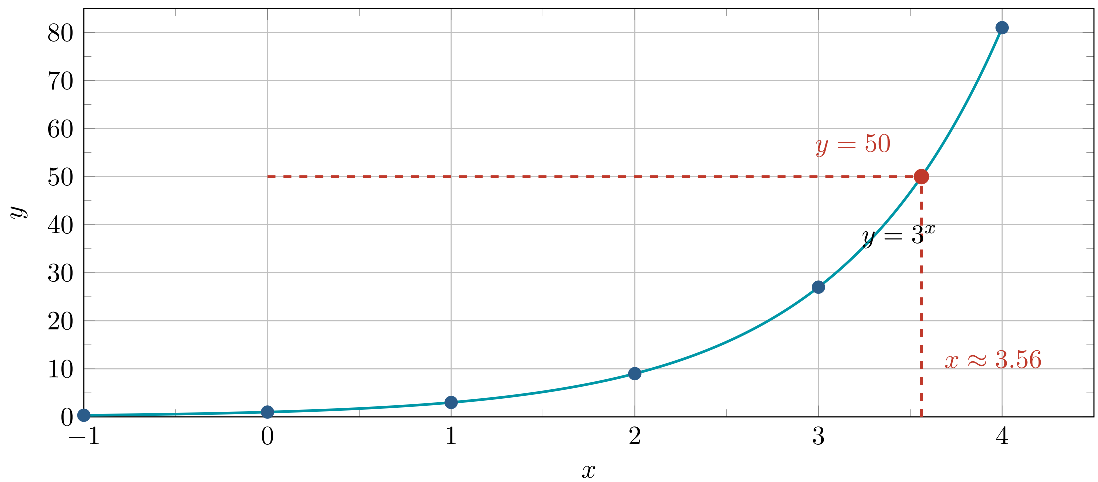
- q15: on the graph, where is $y = 50$? Between which two integer x-values?
- q15: why does the x-value where the graph hits $y = 50$ equal $\log_3 50$?
- q15: $3^3 = 27$, $3^4 = 81$ — so $\log_3 50$ is between ___ and ___?
- q17: what does $\log_3(1/27) = y$ mean in exponential form?
- q52: does the log change or the argument?
- q53: can you rewrite in exponential form first?
- q54: what is the domain of $\log_b x$? What is the asymptote?

Adult role

Solo teacher: 4 laps. On q15, press the 'why does this work?' reasoning — that's the Form B Q10/Q11 assessment-critical move.

Practice #15 [STAR] DOK-3 SPINE (6-3-savvas-q15)

Higher Order Thinking Use the graph of $y = 3^x$ to estimate the value of $\log_3 50$. Explain your reasoning.
[GRAPH / TIKZ figure]



[STAR] DOK-3 SPINE — Practice #15 (exp↔log inverse via graph)

Prompt: Use the graph of $y = 3^x$ to estimate the value of $\log_3 50$. Explain your reasoning.
GRAPH: Student packet has $y = 3^x$ graph ($x \in [-1, 4.5]$, $y \in [0, 85]$). Teacher copy: same graph + dashed $y = 50$ line + read-off annotation. [GRAPH PLACEHOLDER — needs_cleanup=YES]
Expected reasoning: Find $y = 50$ on graph, read x-value. $3^3 = 27$, $3^4 = 81 \to$ answer between 3 and 4, closer to 3.5. Formally $\log_3 50 \approx 3.56$.
Pair-share: why does this work? Because $\log_3 50 = x$ means $3^x = 50$ — the x-value where the exp graph hits $y = 50$ IS the log.
Form B alignment: Q7 (inverse of exp), Q10 (inverse from graph), Q11 (log VA).

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): 3.5; $\log_3 50$ is an expression equivalent to $3^x = 50$, so you need to find the value of x for which $y = 50$.

[SPEAK] CHAIN 2 CALLBACK — Compound Interest (CLOSURE, 3 priors)

When to say: Before students start Practice #52.

"We've done this three times now — polynomial, rational, exponential decay. Today: continuous compound, $y = Pe^{rt}$. The model is **given**, we solve for t — same as always. What's new is that to invert e^t we need a new tool, the natural log. **That's why this unit exists** — to give us the lock-pick for exponential equations. Every time we've done model-then-extract, we needed an inverse operation. Today's inverse is ln. Same move, fifth representation, new tool."

This is the line that makes students feel the unit is earning its keep rather than being a new abstract topic.

Practice #52 (6-3-savvas-q52)
How long does it take for \$250 to grow to \$600 at 4% annual percentage rate compounded continuously? Round to the nearest year.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

Practice #53 (6-3-savvas-q53)

Model with Mathematics Michael invests \$1,000 in an account that earns a 4.75% annual percentage rate compounded continuously. Peter invests \$1,200 in an account that earns a 4.25% annual percentage rate compounded continuously. Which person's account will grow to \$1,800 first?

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

[SPEAK] CHAIN 1 CALLBACK — Richter (representation shift, log family)

When to say: Before students start Practice #54. This is the first appearance of Chain 1 in the log family — worth a pause.

"Here's the move one more time: given a formula tying two quantities together (amplitude A and magnitude R), we want to extract one given the other. The difference: this time the link between them is a logarithm, not a polynomial or a cube root. We're going to spend this lesson learning that logs can live inside the same move we've been practicing since Storage Box."

Practice #54 (6-3-savvas-q54)

Reason The Richter magnitude of an earthquake is $R = 0.67\log(0.37E) + 1.46$, where E is the energy (in kilowatt-hours) released by the earthquake.

[IMAGE: Cracked wall photo with Richter scale overlay numbered 1 through 10 and labels "Not felt" at 1 and "Extreme" near 9–10; callout text: "At a richter magnitude of 4 and above, the walls in your house may start to crack."**]**

a. What is the magnitude of an earthquake that releases 11,800,000,000 kilowatt-hours of energy? Round to the nearest tenth.
b. How many kilowatt-hours of energy would earthquake have to release in order to be an 8.2 on the Richter scale? Round to the nearest whole number.
c. What number of kilowatt-hours of energy would an earthquake have to release in order for walls to crack? Round to the nearest whole number.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. magnitude 7.9 b. 31,009,857,353 kWh c. 16,705 kWh · IMAGE PLACEHOLDER [photo]: Cracked wall photo with Richter scale overlay numbered 1 through 10 and labels "Not felt" at 1 and "Extreme" near 9–10; callout text: "At a richter magnitude of 4 and above, the walls in your house may start to crack."

Practice #17 (promoted to core) (6-3-savvas-q17)

Use Structure A student says that $\log_3\left(\frac{1}{27}\right)$ simplifies to -3. Is the student correct? Explain.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

[PUZZLE] FORM B ASSESSMENT ALIGNMENT

- q15 (spine) → Form B Q7 (inverse of exp), Q10 (inverse from graph), Q11 (log VA)
- q17 → Form B Q13 (condense argument reasoning)
- q52 → Form B Q9 (log↔exp definition)
- q53 → Form B Q14 (solve exp with logs, Change of Base)
- q54 → Form B Q9 + Q14 (multi-part log modeling)

Tag: academic conversation + 6-4 bridge

Share / Summary **DOK:** 2 **Minutes:** 5

Teacher does

- Cold-call 1 pair to present q15 reasoning using sentence frame.
- Class signals thumbs. Write on board: $\log_3 50 = x$ means $3^x = 50$.
- 6-4 bridge: 'Next lesson: logarithmic FUNCTIONS — graphing $y = \log_b x$. Today's log ↔ exp definition is the foundation.'

Students do

- Presenting pair explains: 'I found $y = 50$ on the graph and read $x \approx 3.56$. This works because $\log_3 50 = x$ means $3^x = 50$.'
- Rest of class signals and writes takeaway in margin.

Questions to ask

- Summarize: how does the graph of $y = 3^x$ let you find $\log_3 50$?
- Why is it that reading x gives you the log? What is the relationship?

Adult role

Cold-call a pair that got the estimation — hold them accountable to naming the exp↔log inverse relationship explicitly.

Tag: summary recap (not graded)

Exit Ticket **DOK:** 1–2 **Minutes:** 5

Teacher does

- Collect at door.
- Scan blank 3 ($e^t = 6.125 \to \ln$). Plan re-teach if many students write 'log' instead of 'ln' or leave the base blank.

Students do

- 4 sentence stems, honest.

Questions to ask

- Did students correctly state $\log_b x = y$ means $b^y = x$?
- Did students get $\log_2 32 = 5$ (since $2^5 = 32$)?
- Did students write ln (not log) for base e?

Adult role

Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

$\log_b x = y$ means _____.
$\log_2 32 =$ _____ because $2^? = 32$.
To solve $e^t = 6.125$, take the _____ of both sides giving $t =$ _____.
One biggest thing I learned today: ____.

[CLOCK] PERIOD A / PERIOD F — 50-MIN CLASS NOTES

Base lesson is 50 min. If 65 min: use extra 15 min on Explore q15 (richer pair-share on why exp↔log inversion works) and q53/q54.
45-min Wed F variant: cut q53 + q54 from Explore. NEVER cut q15.
If finished early: hand out Practice #18 ($\ln 1000 \neq 3$ — fast finisher).

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Log ↔ Exp Rules card as a sticker/cut-out.
- For q15, pre-label the y-axis at y=27, y=50, y=81 so students can locate the target values without visual confusion.
- Accept 'between 3 and 4, closer to 3.5' as full credit for the estimate.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: 'logarithm' = the exponent; 'inverse' = undoes; 'estimate' = approximate value, not exact.
- 'UNDEFINED' = no real number answer (not zero).
- Pair ELL students with English-dominant partners for TPS.